AI SYSTEM BRIEF & ARCHITECTURE SPECIFICATION

Automated Support Ticket Triage Workflow

Platform: n8n Orchestration | **Pattern:** Deterministic Guardrails + LLM Classification + HITL | **Repository:** `ticket-triage-automation/`

1. Executive Overview

The **Automated Support Ticket Triage Workflow** is an intelligent customer operations pipeline built in n8n. It combines deterministic pre-filtering rules with structured LLM extraction and classification (Google Gemini) to evaluate inbound customer tickets, escalate high-risk inquiries, log audit trails, route standard issues, and incorporate Human-in-the-Loop (HITL) approval before dispatching automated customer responses.

2. Core Business Problem

> **Operational Bottleneck:** Manual ticket triage slows down response times during high-volume periods, risks delayed intervention on urgent or sensitive issues, and exposes teams to brand risk if unverified AI responses are sent directly to customers without oversight.

3. Architecture & Execution Flow

Stage	n8n Node / Component	Functional Responsibility
1. Ingestion & Prep	`Webhook` → `Prepare`	Receives inbound ticket payloads via POST request and normalizes key metadata fields.
2. Deterministic Guardrails	`High risk keyword check` & `Missing order id check`	Immediately flags sensitive keywords or missing order identifiers and triggers instant Slack escalation alerts.
3. AI Extraction & Triage	`Extract` & `Classify` (Gemini + Structured Output Parsers)	Parses unstructured ticket text into validated JSON fields and assigns category and priority scores.
4. Audit & Routing	`If node`, `Audit` (Airtable), & `Switch`	Logs structured classification records to an audit database and routes tickets across three resolution paths.
5. HITL Drafting & Approval	`Basic LLM Chain` → `Send message and wait for response` (Slack)	Drafts a tailored customer reply and pauses execution until a human agent approves or rejects the draft in Slack.
6. Final Dispatch	`If` → `Send a message` (Gmail / Slack)	Sends the approved email response to the customer or routes rejected drafts back to support channels for manual handling.

4. Key Architectural Design Principles

- **Rule-First Safety Filtering:** Critical risk checks execute deterministically before any LLM inference, ensuring zero latency on urgent escalations and saving unnecessary token costs.
- **Schema-Enforced AI Outputs:** Dedicated Structured Output Parsers guarantee predictable JSON structures across extraction, classification, and response drafting stages.
- **Human-in-the-Loop Governance:** Built-in Slack approval gates prevent hallucinated or off-tone AI responses from reaching customers autonomously.